



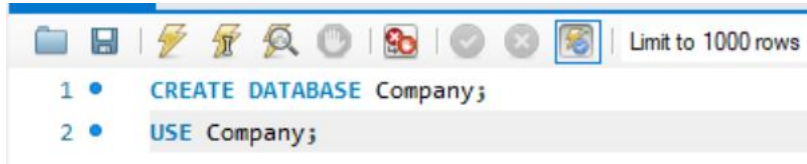
Lab Sheet 1

Database Management System Lab (BTCS203P)

B. TECH AI&ML, CYBER SECURITY, Data Science 3RD Semester (Session 2026-2025)

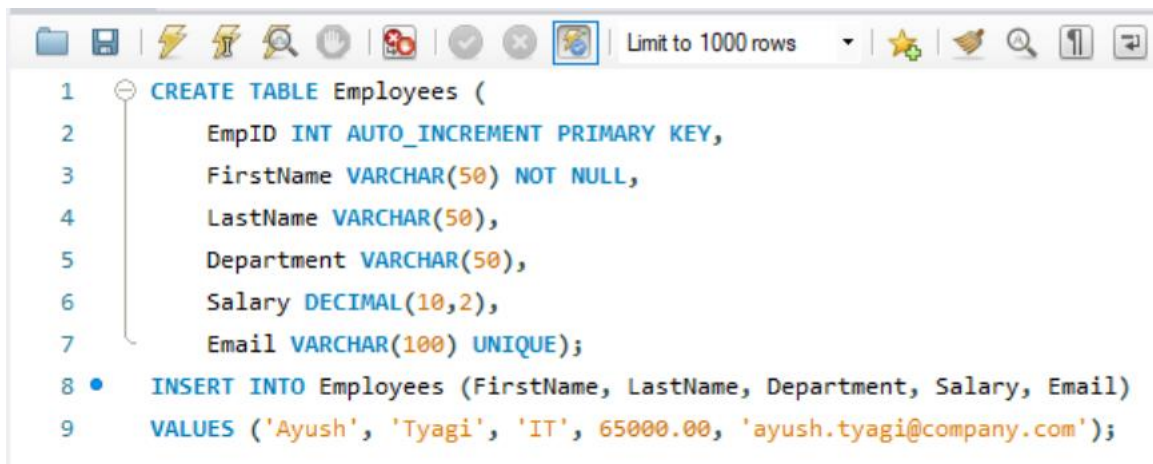
Task 1: Create a Table

a. Create a database named Company.



```
1 • CREATE DATABASE Company;
2 • USE Company;
```

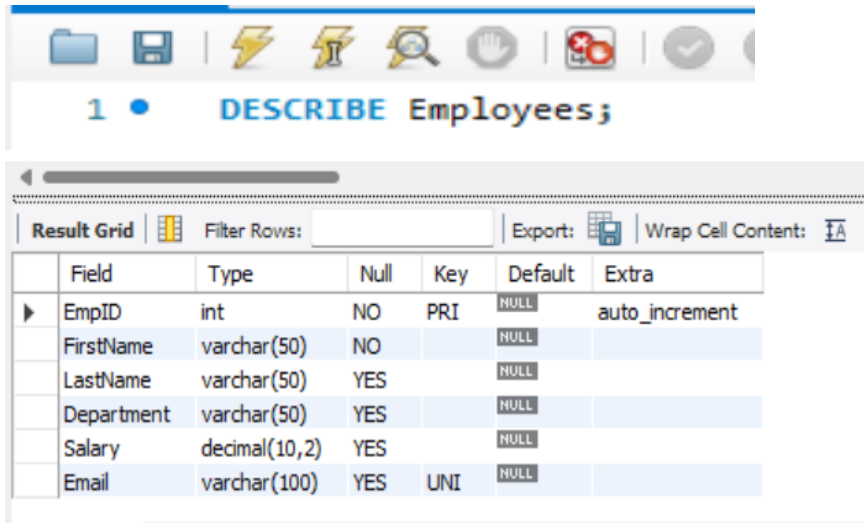
b. Create a table named Employees with the following structure:



```
1 CREATE TABLE Employees (
2     EmpID INT AUTO_INCREMENT PRIMARY KEY,
3     FirstName VARCHAR(50) NOT NULL,
4     LastName VARCHAR(50),
5     Department VARCHAR(50),
6     Salary DECIMAL(10,2),
7     Email VARCHAR(100) UNIQUE);
8 • INSERT INTO Employees (FirstName, LastName, Department, Salary, Email)
9     VALUES ('Ayush', 'Tyagi', 'IT', 65000.00, 'ayush.tyagi@company.com');
```

Task 2: DDL Queries on the Employees Table

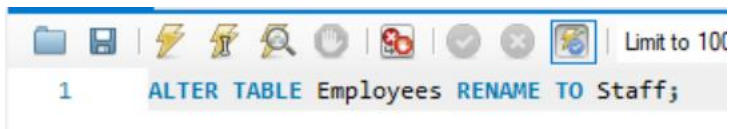
a) Describe the structure of the table.



The screenshot shows a SQL IDE interface. The query editor contains the command `DESCRIBE Employees;`. Below the editor, the 'Result Grid' tab is active, displaying the table structure of 'Employees'.

	Field	Type	Null	Key	Default	Extra
▶	EmpID	int	NO	PRI	NULL	auto_increment
	FirstName	varchar(50)	NO		NULL	
	LastName	varchar(50)	YES		NULL	
	Department	varchar(50)	YES		NULL	
	Salary	decimal(10,2)	YES		NULL	
	Email	varchar(100)	YES	UNI	NULL	

b) Rename the table to Staff.



The screenshot shows a SQL IDE interface with the query editor containing the command `ALTER TABLE Employees RENAME TO Staff;`.

c) Add a new column of Phone Number (VARCHAR (15)).



The screenshot shows a SQL IDE interface with the query editor containing the command `ALTER TABLE Staff ADD COLUMN PhoneNumber VARCHAR(15);`.

d) Modify the data type of Salary to DECIMAL(12,2).



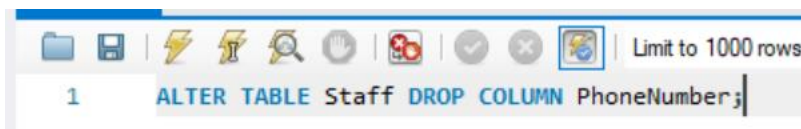
The screenshot shows a SQL IDE interface with the query editor containing the command `ALTER TABLE Staff MODIFY COLUMN Salary DECIMAL(12,2);`.

e) Add a DEFAULT value of 'General' to the Department column.



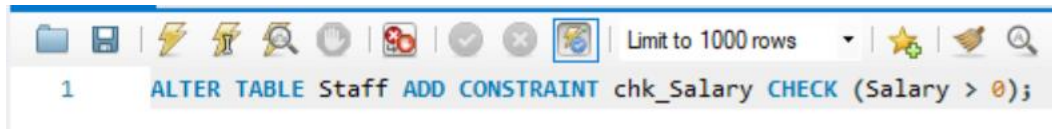
The screenshot shows a SQL IDE interface with the query editor containing the command `ALTER TABLE Staff ALTER COLUMN Department SET DEFAULT 'General';`.

f) Drop the PhoneNumber column.



The screenshot shows a SQL IDE interface with the query editor containing the command `ALTER TABLE Staff DROP COLUMN PhoneNumber;`.

g) Add a CHECK constraint to ensure Salary is greater than 0.



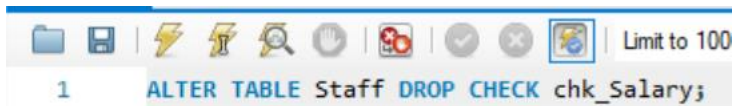
1 ALTER TABLE Staff ADD CONSTRAINT chk_Salary CHECK (Salary > 0);

h) Rename the column FirstName to GivenName.



1 ALTER TABLE Staff CHANGE COLUMN FirstName GivenName VARCHAR(50) NOT NULL;

i) Drop the chk_Salary constraint:



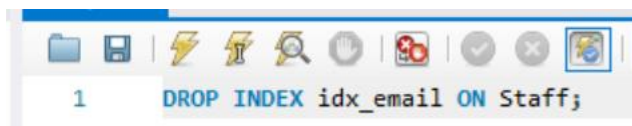
1 ALTER TABLE Staff DROP CHECK chk_Salary;

j) Create an index on the Email column.



1 CREATE INDEX idx_email ON Staff(Email);

k) Drop the index on the Email column.



1 DROP INDEX idx_email ON Staff;

j) Truncate the table to remove all rows without affecting the structure.



1 TRUNCATE TABLE Staff;

k) Drop the Staff table completely.



1 DROP TABLE Staff;

1. Conceptual Questions:

1. What is the difference between TRUNCATE and DELETE?

TRUNCATE is a DDL command that removes all rows from a table by deallocating the data pages, making it faster and non-filterable. DELETE is a DML command that removes rows one by one, logs each deletion, and supports the WHERE clause

2. Explain the use of the CHECK constraint.

The CHECK constraint enforces domain integrity by ensuring that all values placed in a column satisfy a specific logical condition (e.g., Salary > 0).

3. Why is it important to define a PRIMARY KEY for a table?

A PRIMARY KEY uniquely identifies each record in a table, enforces entity integrity by preventing duplicate or NULL entries, and automatically creates a clustered index for optimized data retrieval.

4. How does ALTER TABLE help in modifying an existing table?

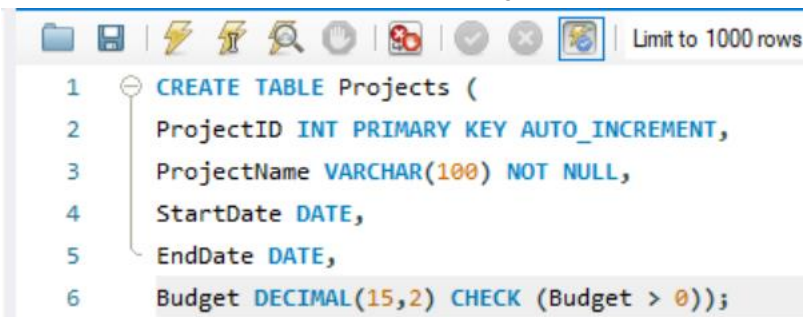
The ALTER TABLE statement allows structural changes to an existing database table (such as adding, modifying, or dropping columns and constraints) without losing existing data.

5. What is the purpose of creating an index in a table?

An index speeds up the data retrieval process (SELECT queries) by creating a pointer structure to rows, reducing the need for full table scans.

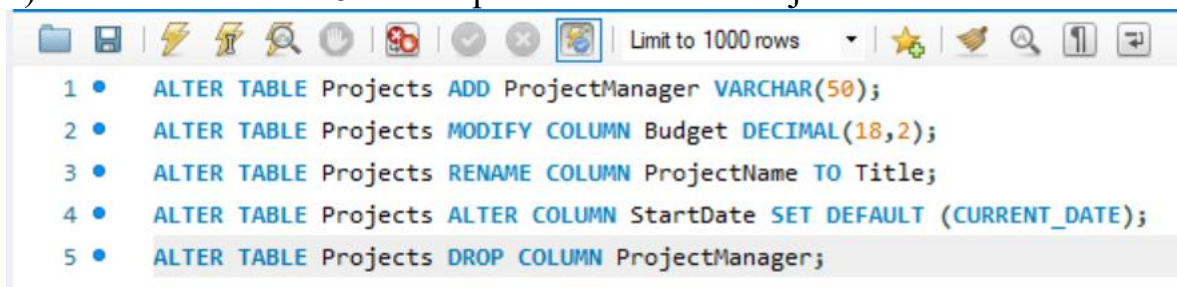
2. Exercises:

1. Create a table named Projects with the following structure:



```
1 CREATE TABLE Projects (  
2   ProjectID INT PRIMARY KEY AUTO_INCREMENT,  
3   ProjectName VARCHAR(100) NOT NULL,  
4   StartDate DATE,  
5   EndDate DATE,  
6   Budget DECIMAL(15,2) CHECK (Budget > 0));
```

a) Perform at least 5 DDL operations on the Projects table.



```
1 ALTER TABLE Projects ADD ProjectManager VARCHAR(50);  
2 ALTER TABLE Projects MODIFY COLUMN Budget DECIMAL(18,2);  
3 ALTER TABLE Projects RENAME COLUMN ProjectName TO Title;  
4 ALTER TABLE Projects ALTER COLUMN StartDate SET DEFAULT (CURRENT_DATE);  
5 ALTER TABLE Projects DROP COLUMN ProjectManager;
```

b) Write a query to retrieve the names of all columns in the Staff table.



```
1 • SELECT COLUMN_NAME FROM INFORMATION_SCHEMA.COLUMNS
2   WHERE TABLE_SCHEMA = 'Company' AND TABLE_NAME= 'Staff';
```

c) Write a query to modify the Department column to allow NULL values.



```
1 • ALTER TABLE Staff MODIFY COLUMN Department VARCHAR(50) NULL;
```

